

## Lab 20: PAC Learning / Sample Complexity

### Aim

To calculate the **sample complexity** required in **PAC (Probably Approximately Correct) Learning** for given accuracy and confidence values.

### Algorithm

1. Define the hypothesis space size  $H$ .
2. Define the error tolerance  $\epsilon$ .
3. Define the confidence parameter  $\delta$ .
4. Apply the PAC sample complexity formula.
5. Calculate the minimum number of training examples required.
6. Display the result.

### Program

```
# Lab 20: PAC Learning / Sample Complexity
```

```
import math
```

```
# Number of hypotheses
```

```
H = 100
```

```
# Error tolerance
```

```
epsilon = 0.1
```

```
# Confidence parameter
```

```
delta = 0.05
```

```
# PAC sample complexity formula
```

```
m = (1 / epsilon) * (  
    math.log(H) + math.log(1 / delta)  
)  
  
print("Minimum number of training examples:", math.ceil(m))
```

### **Sample Output**

Minimum number of training examples: 77

### **Result**

Thus, the **PAC Learning sample complexity was successfully calculated**, and the minimum number of training examples required for the given accuracy and confidence values was obtained.